

Pre-Board Examination – 2079

Class : XII

Subject: C. English (0041)

Full Marks: 75

Time: 3 hrs.

SET A

Pass Marks: 30

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

1. Read the passage given below and answer the questions that follow.

- A.** The Scottish bacteriologist Dr Alexander Fleming (1881-1955) is credited with the discovery of penicillin in London in 1928. He had been working at St Mary's Hospital on the bacteriology of septic wounds. As a medic during World War I, he had witnessed the deaths of many wounded soldiers from infection and he had observed that the use of harsh antiseptics, rather than healing the body, actually harmed the blood corpuscles that destroy bacteria.
- B.** In his search for effective antimicrobial agents, Fleming was cultivating staphylococcus bacteria in Petri dishes containing agar*. Before going on holiday in the summer of 1928, he piled up the agar plates to make room for someone else to use his workbench in his absence and left the windows open. When he returned to work two weeks later, Fleming noticed mould growing on those culture plates that had not been fully immersed in sterilising agent. This was not an unusual phenomenon, except in this case the particular mould seemed to have killed the staphylococcus aureus immediately surrounding it. He realised that this mould had potential. (*agar is a culture medium based on a seaweed extract – used for growing microorganisms in laboratories)
- C.** Fleming consulted a mycologist called C J La Touche, who occupied a laboratory downstairs containing many mould specimens (possibly the source of the original contamination), and they concluded it was the *Penicillium* genus of ascomycetous fungi. Fleming continued to experiment with the mould on other pathogenic bacteria, finding that it successfully killed a large number of

them. Importantly, it was also non-toxic, so here was a bacteria-destroying agent that could be used as an antiseptic in wounds without damaging the human body. However, he was unsuccessful in his attempts to isolate the active antibacterial element, which he called penicillin. In 1929, he wrote a paper on his findings, published in the *British Journal of Experimental Pathology*, but it failed to kindle any interest at the time.

- D.** In 1938, Dr Howard Florey, a professor of pathology at Oxford University, came across Fleming's paper. In collaboration with his colleague Dr Ernst Chain, and other skilled chemists, he worked on producing a usable drug. They experimented on mice infected with streptococcus. Those untreated died, while those injected with penicillin survived. It was time to test the drug on humans but they could not produce enough – it took 2,000 litres of mould culture fluid to acquire enough penicillin to treat a single patient. Their first case in 1940, an Oxford police officer who was near death as a result of infection by both staphylococci and streptococci, rallied after five days of treatment but, when the supply of penicillin ran out, he eventually died.
- E.** In 1941, Florey and biochemist Dr Norman Heatley went to the United States to team up with American scientists with a view to finding a way of making large quantities of the drug. It became obvious that *Penicillium notatum* would never generate enough penicillin for effective treatments so they began to look for a more productive species. One day a laboratory assistant turned up with a melon covered in mould. This fungus was *Penicillium chrysogenum*, which produced 200 times more penicillin than Fleming's original species but, with further enhancement and filtration, it was induced to yield 1,000 times as much as *Penicillium notatum*. Manufacture could begin in earnest.
- F.** The standardisation and large-scale production of the penicillin drug during World War II and its availability for treating wounded soldiers undoubtedly saved many lives. Penicillin proved to be very effective in the treatment of pneumococcal pneumonia – the death rate in WWII was 1% compared to 18% in WWI. It has since proved its worth in the treatment of many life-threatening infections such as tuberculosis, meningitis, diphtheria and several sexually-transmitted diseases.

- G.** Fleming has always been acknowledged as the discoverer of penicillin. However, the development of a commercial penicillin drug was due to the skill of chemical scientists Florey, Chain and others who overcame the difficulties of converting it into a usable form. Fleming and Florey received knighthoods in 1944 and they, together with Chain, were awarded the Nobel Prize in Physiology or Medicine in 1945. Heatley's contribution seems to have been overlooked until, in 1990, he was awarded an honorary doctorate of medicine by Oxford University – the first in its 800-year history.
- H.** Fleming was mindful of the dangers of resistance to penicillin early on and he expressly warned on many occasions against overuse of the drug, because this would lead to bacterial resistance. Ironically, the occurrence of resistance is pushing the drive today to find new, more powerful antibiotics.
- A.** Based on your understanding of the above passage, match the statements given below with the help of eight paragraphs, **A–H**. Which paragraph contains the following information? Write the correct letter, **A–H**, in boxes 1–5 on your answer sheet. **[5×1 = 5]**
- a) results of animal experiments
 - b) recognition of the scientists' valuable work
 - c) a statement about the beginning of mass production
 - d) Fleming's cautionary advice
 - e) examples of uses for penicillin
 - f) the starting point for Fleming's original research
- B.** Complete the summary below. Choose *NO MORE THAN TWO WORDS* from the passage for each answer. **[5×1 = 5]**

Dr Fleming's Accidental Discovery

In a bid to find a safe and effective antiseptic, Dr Fleming was growing (a)bacteria in his lab. On his return from (b), he found mould on an unsterilised plate and saw that it had destroyed the bacteria around it. A (c) helped him identify the mould. Fleming found that it was active against several different (d) and, because it was (e), it was safe to use in humans.

- C. Choose the corresponding **years** from the passage in relation to the details of actions as per the timeline events below: [5×1 = 5]

Timeline Events

Example Q:: Fleming’s discovery of Penicillin

Answer: **1928**

- a): Fleming’s research published
- b): Florey begins work on penicillin
- c): The first human subject (eventually) died
- d): Collaboration with American scientists
- e): Heatley’s work is acknowledged

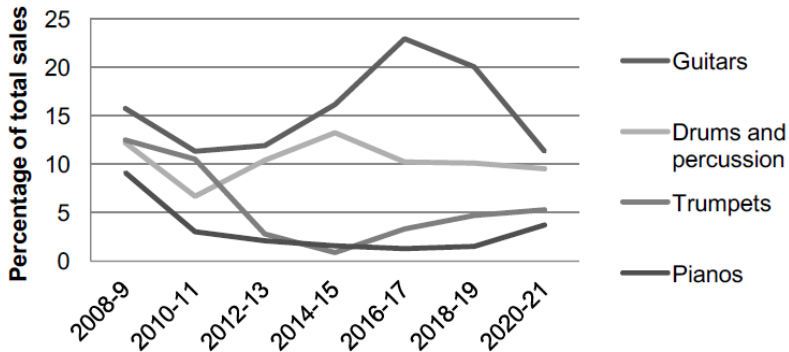
2. **Write short answers to the following questions.** [5×2 = 10]

- (a) Discuss the title’s significance in the poem, *Soft Storm*.
- (b) What is the central idea of the poem, *The Awakening Age*?
- (c) Critically examine the relationship between Monsieur Duran and his wife. (*Facing Death*)
- (d) “You are not the center of the world” Discuss the statement with reference to the essay, *Humility*.
- (e) What is the significance of another human rights movement according to Samuel Moyn? Discuss (*Human Rights and the Age of Inequality*)

3. **Write long answers to the following questions.** [2×5 = 10]

- (a) “On Libraries” is written in praise of intellectual freedom, community work, and the ecstasy of serendipitous discovery.” Discuss.
- (b) Discuss the six dimensions that define normative marriage in the United States of America. (*Marriage as a Social Institution*)

4. The graph below shows the changes in the sales of musical instruments in Nepal from 2008 to 2021. Summarise the information by selecting and reporting the main features, and make comparisons where relevant. [7]



5. Write a letter to the customer service manager of New Home Solutions, Pvt. Ltd., New Baneshwor, complaining about a defect in the music system purchased from the store. Sign yourself as Ram Nepal. (180 words) [8]
6. The internet has changed the way we communicate. Much communication today happens through social media. Some people support this and think it is a positive development. Others believe that social media have negative effects. Discuss both these views and give your own opinion. (300 words) [10]
7. **Do as directed and rewrite the sentences.** [10×1 = 10]
- (a) I (must/should/could) have checked the parking restrictions before I left my car there! Now I'll have to pay a fine! (*Choose the best modal verb.*)
- (b) Could you show me where the snack bar is? (*Convert into a direct question.*)
- (c) 'Who did you see?' she asked. (*Change it into indirect speech.*)

- (d) You heat ice. It melts. (*Convert the sentences into **if sentence**.*)
- (e) The documents/the travelers/will distribute/ the air hostess (*Make a passive sentence with the information.*)
- (f) We were standing outside the cinema. Someone picked my pocket (*Join the sentences with either **while** or **when**.*)
- (g) Are you the one? Did you send me the email? (*Join the sentences with a suitable relative clause.*)
- (h) Stay disciplined and be patient, your time..... (*Fill in the blanks with a suitable verb of **come**.*)
- (i) These shoes cost me 50 dollars. (*Fill in the blanks with a suitable preposition: **in/on/around/above**.*)
- (j) When I was a child, we live in the countryside. (*Rewrite the sentence with either **used to** or **would**.*)

8. Do as directed.

[5×1 = 5]

- (a) Write the adverb form of the word **rationality**.
- (b) The word '**knowledge**' has the correct IPA phonetic inscription:
 - i. /'nɒl.ɪdʒ/ ii. /'nɒl.ɪz/ iii. /'nɛl.ɪdʒ/ iv. /'nɒl.ɪdʒ/
- (c) How many syllables are in the word: **supercelestial**?
- (d) She is a (Nepali, beautiful, tall, black-haired) woman. (*Arrange the adjectives in the correct order to make a sensible sentence.*)
- (e) They have been working hard. (*Copy the sentence and underline the content word(s)*)

❧ - The End - ❧